CS & IT ENGINEERING

COMPILER DESIGN

SYNTAX DIRECTED TRANSLATION

Lecture No.- 02

By- Venkat sir

Recap of Previous Lecture

Topic Semantic Analysis

Topic Syntax Directed translation Semantic errors

Topics to be Covered

Topic Syntax Directed translation Construction

Topic ?? Synthesized attribute

Topic ?? Inherited Attribute

S-attributed Definition

L-attributed Definition

Translation Scheme

Topic : Syntax Directed Translation

Steps to construct SDT:-

(1) Construct Parse tree.

(2) Construct annotated parse tree

(3) Compute attribute information

(4) Generalize rules and attach to grammar production.

Topic : Syntax Directed Translation

Construct SDT to Count total no. of balanced parenthesy present in the input

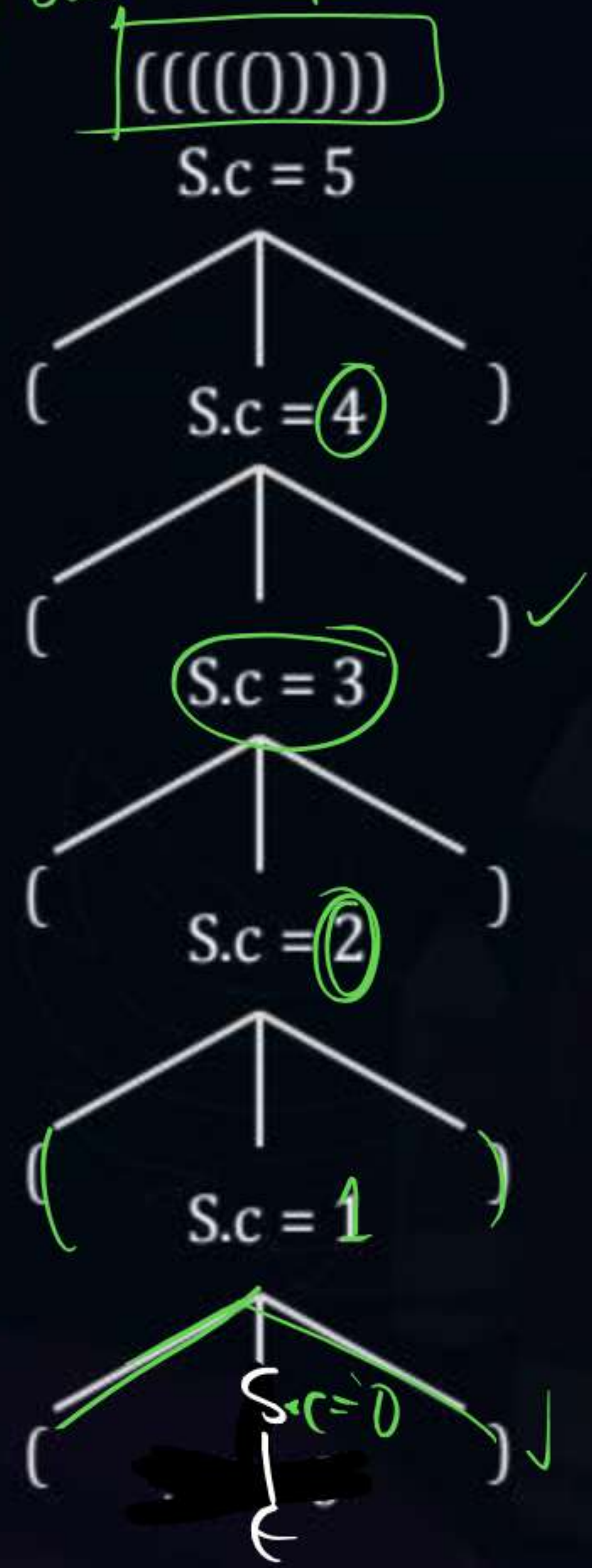
#Q. $S \to (S_1)$ $\{S.count = S_1.count + 1\}$

$S \to \epsilon$ $\{S.count = 0\}$

Balanced Parenthesis

Output for input

((((()))))



(1) Parse tree

(2) Annotated PT

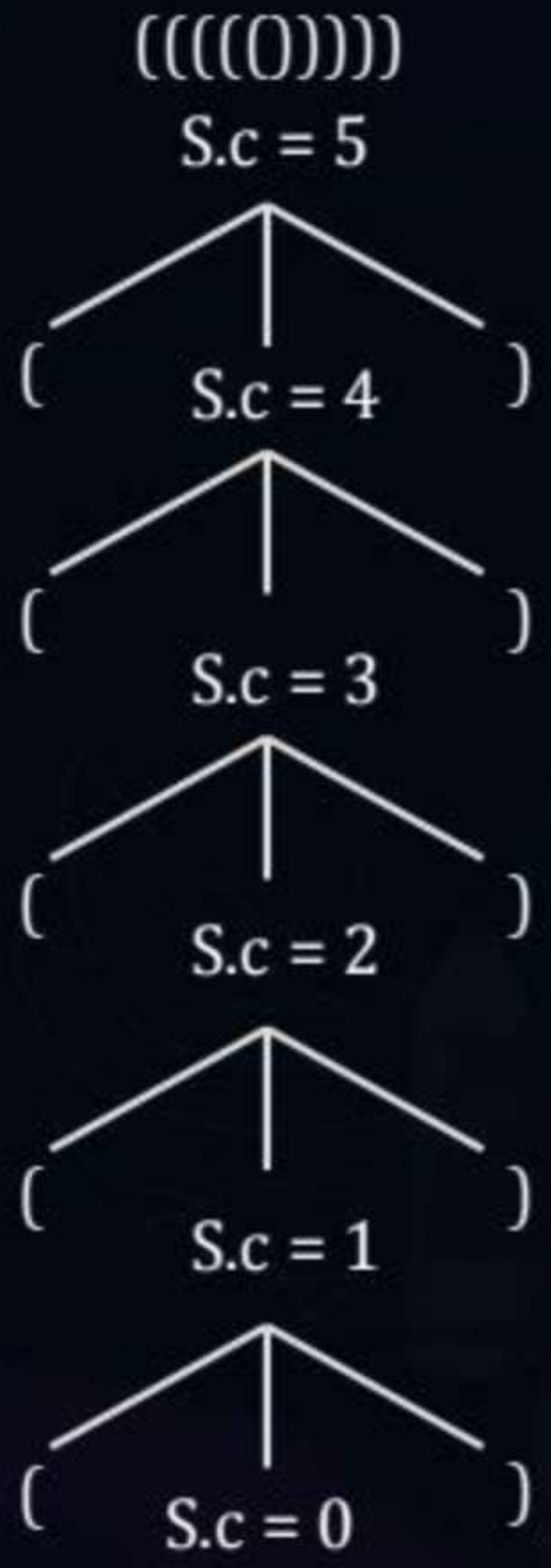
Topic : Syntax Directed Translation

#Q. $S \rightarrow (S_1)$ $\quad\{S.c = S1.\text{count} + 1\}$

$S \rightarrow \in$ $\quad\{S.\text{count} = 0\}$

Balanced Parenthesis

Output for input



(1) Parse tree

(2) Annotated PT

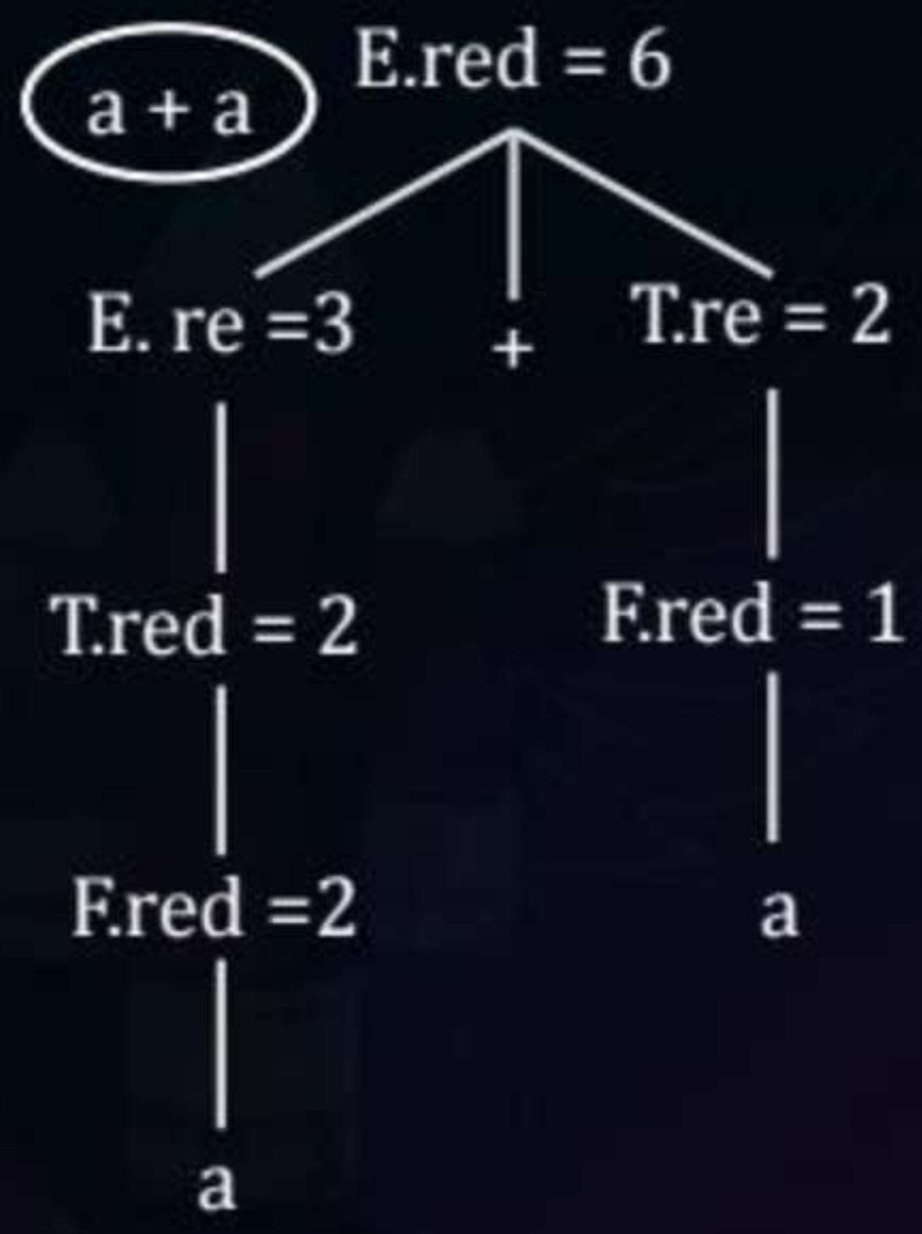
Topic : Syntax Directed Translation

#Q. Construct SDT to find total no of reduce actions taken by B. U. P

1. $E \rightarrow E + T$ {E. reduce = E1. reduce + T. reduce + 1}
2. $E \rightarrow T$ {E reduce = T. red + 1}
3. $T \rightarrow T1 * F$ {T.reduce = T1. reduce + F.reduce + 1}
4. $T \rightarrow F$ {T.reduce = F.reduce + 1}
5. $F \rightarrow a$ {F. reduce = 1}

E. reduce = ?

Annotated peresters



Topic : Syntax Directed Translation

Synthesized

#Q. Construct SDT to find height of the parse tree (height)

$E \rightarrow E_1 + T$ {E. height = max(E1.height, T.height) + 1}

$E \rightarrow T$ {E. height = T. height + 1}

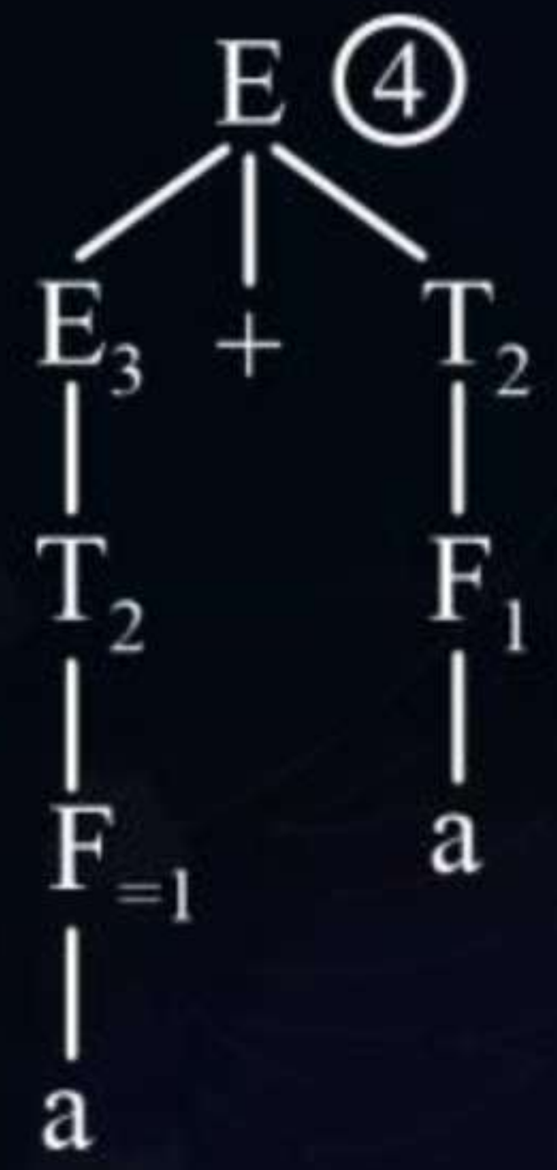
$T \rightarrow T_1 * F$ {T. height = max (T. height, F.height) + 1 }

$T \rightarrow F$ {T. height = F.height + 1}

$F \rightarrow a$ {F.height = 1}

Input 4 ← a + a

What output this SDT produce ?



(Q) Construct SDT to find height of the Parse tree using given grammar

E.height

$E \rightarrow E_1 + T$ {E.height = }

$E \rightarrow T$ {E.height = }

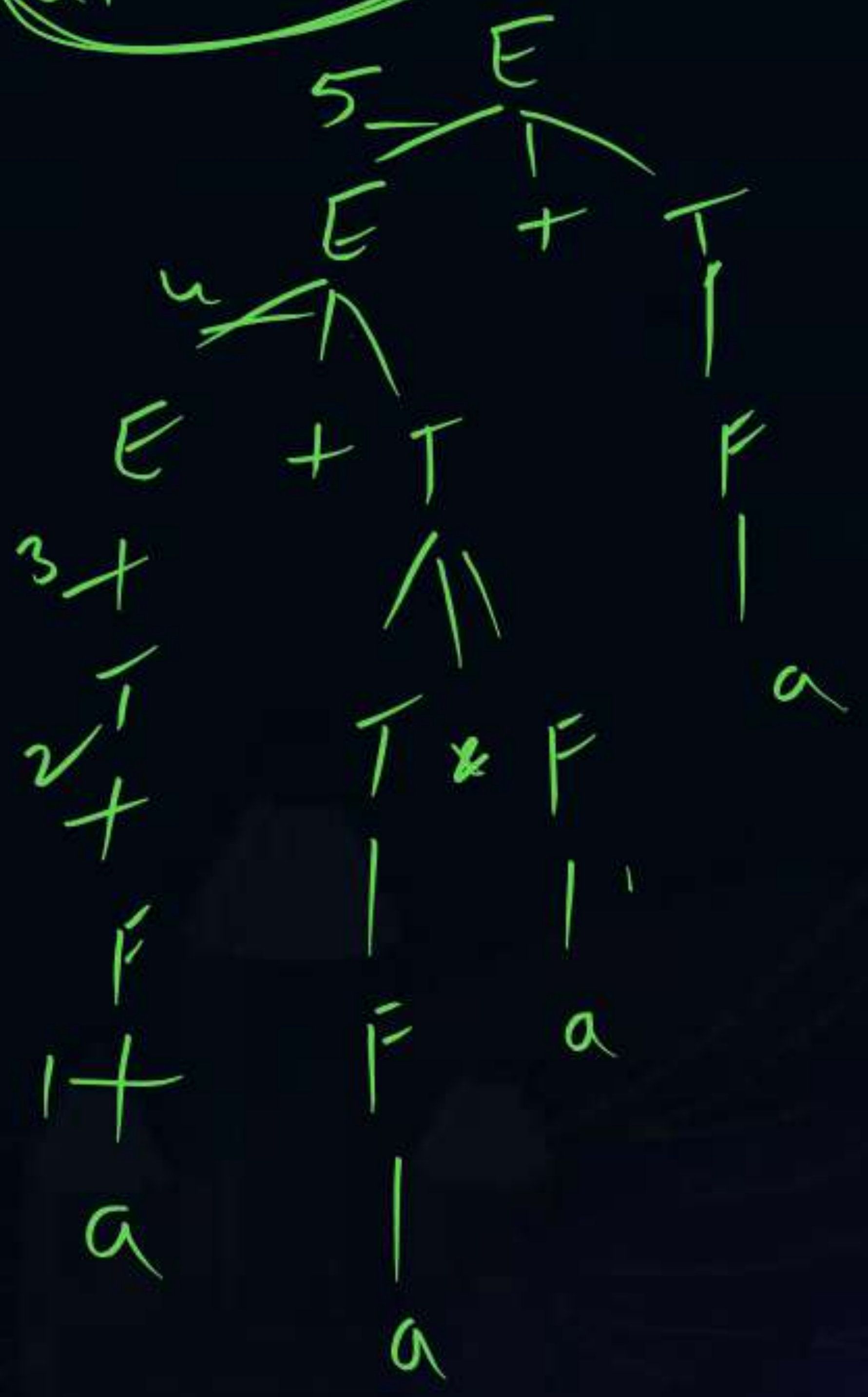
$T \rightarrow T_1 * F$ {T.height = }

$T \rightarrow F$ {T.height = F.h + 1}

$F \rightarrow a$ {F.height = 1}

a + a

a+a*a+a



Topic : Syntax Directed Translation

S attributed Definition

$\{1011\} \Rightarrow 3$ ✓

$101011 \Rightarrow 4$ ✓

#Q. Construct SDT to find total no. of 1's present in the binary string using following grammar

$S \rightarrow L$ $\{S.count = L.count\}$

$L \rightarrow L_1 B$ $\{L.count = L_1.count + B.count\} \Rightarrow$ Synthesized

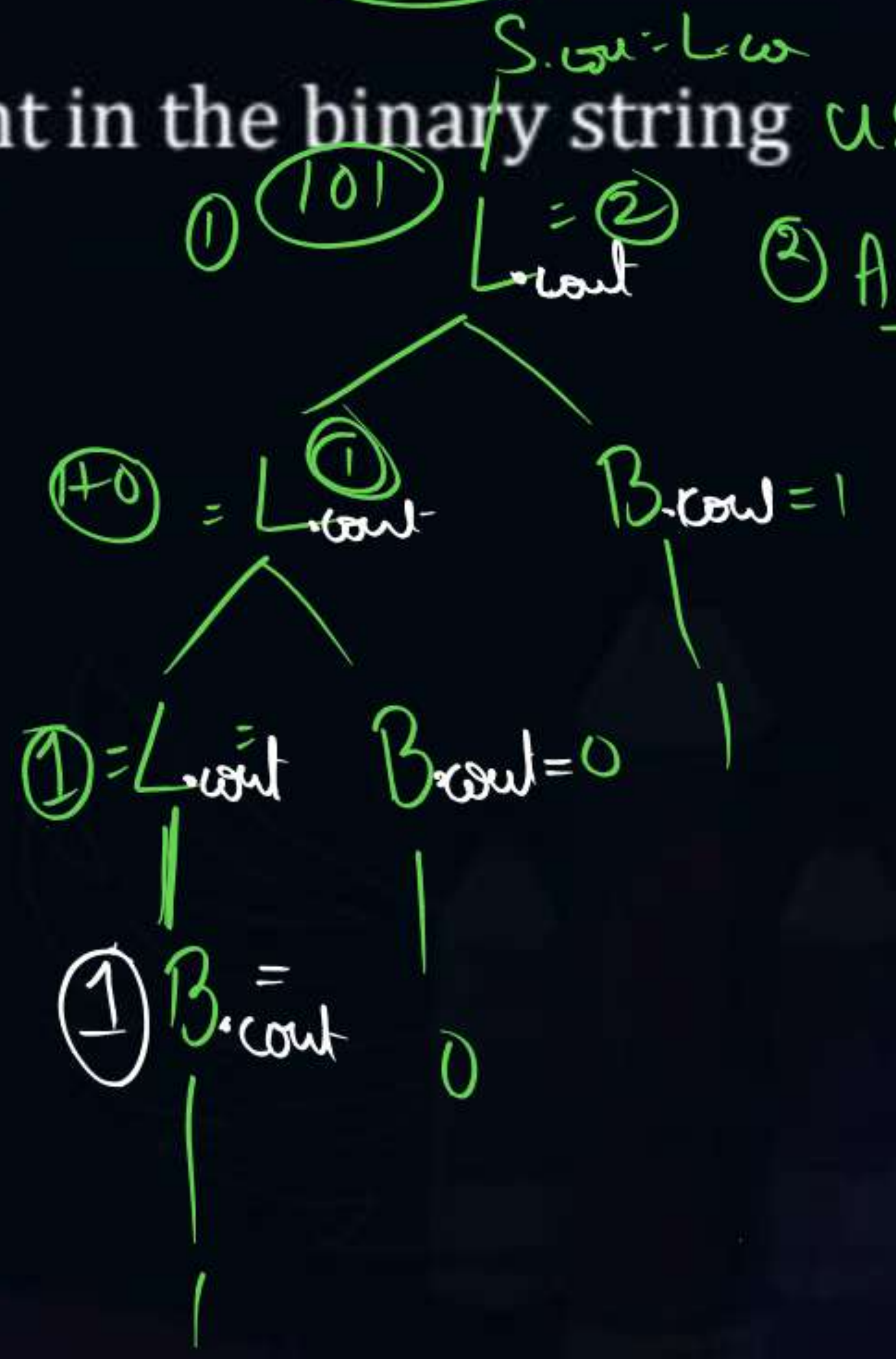
$L \rightarrow B$ $\{L.count = B.count\} \Rightarrow$ Synthesized

$B \rightarrow 0$ $\{B.count = 0\}$

$B \rightarrow 1$ $\{B.count = 1\}$

① 101

② Annotated Parse tree



Topic : Syntax Directed Translation

int a, b ;

int a, b ;

L-attributed Definition

#Q. Construct SDT to fill type information in the symbol table.

S → TL ; {L.itype = T.stype} ① Inherited

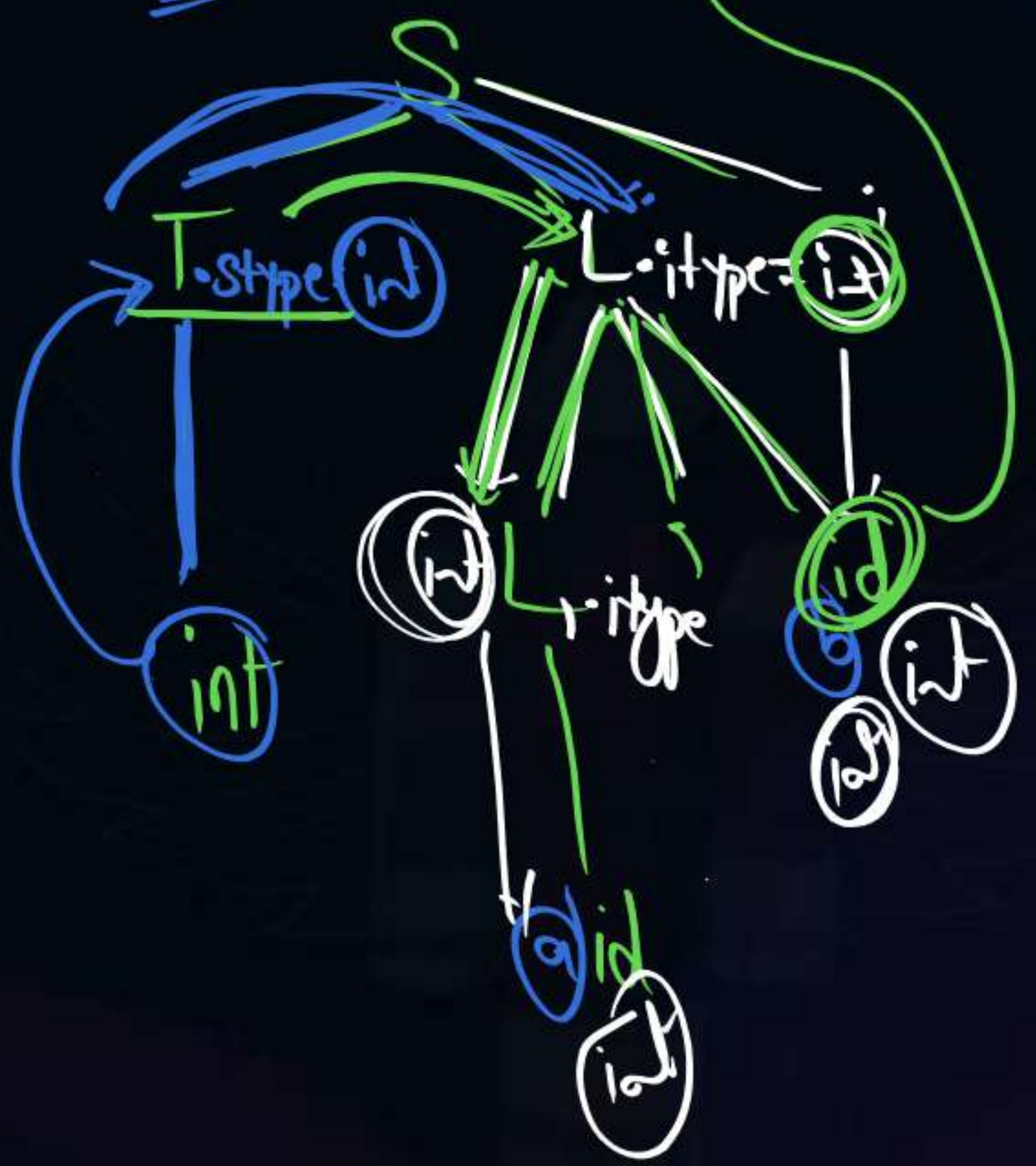
T → int {T.stype = int} ② Synthesized

int L → L, id {L_1.itype = L.itype} ③ Inherited
{addtype("id.name", L.itype)}

④ L → id {addtype("id.name", L.itype)}

① Parse tree

		type
a	id	int
b	id	int



Topic : Syntax Directed Translation

$A \rightarrow BC$

$A = BC$

Synthesized Attribute:-

Attribute value at a node is computed from its children.

$\underline{A} \rightarrow BC$

Topic : Syntax Directed Translation

Inherited Attribute:-

Attribute value at a node computed from parent (or) left sibling.

$A \rightarrow BC$

1. $\{C.x = A.x\}$ ✓
2. $\{C.x = B.c\}$ ✓
3. $\{B.x = C.x\}$ ✗

Topic : Syntax Directed Translation

(SDD)

S – attributed Definition:-

① only Synthesized attributes used

L – attributed Definition:-

① Both Synthesized and Inherited are used

Topic : Syntax Directed Translation

Consider the following SDT

1. S→ aA- {print (1)}
2. S→ a- {print (2)}
3. S→ Sb- {print (3)}

This SDT is carried out with BUP what is output for input abb.

Rule   Semantic

231   B.U.P

Output

S

a A

Sab   a S b 2

aab   a a b 1

Topic : Syntax Directed Translation

Synthesized Attribute

Attribute value at a node is compute form its children $x \rightarrow yz$

$S \rightarrow AB \qquad \{S.x = A.x + B.x\}$

$A \rightarrow a \qquad \{A.x = a\}$

$B \rightarrow b$

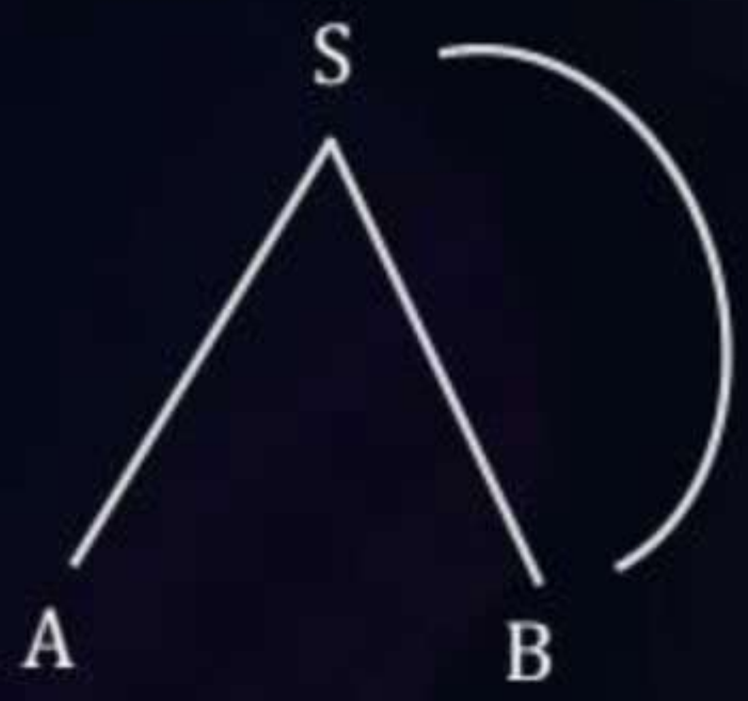
Topic : Syntax Directed Translation

Inherited Attribute:

Attribute value at a node compute from parent (or) left sibling

$S \to AB \{A.x = S.x\}$

$A \to a \{A.x = a\}$

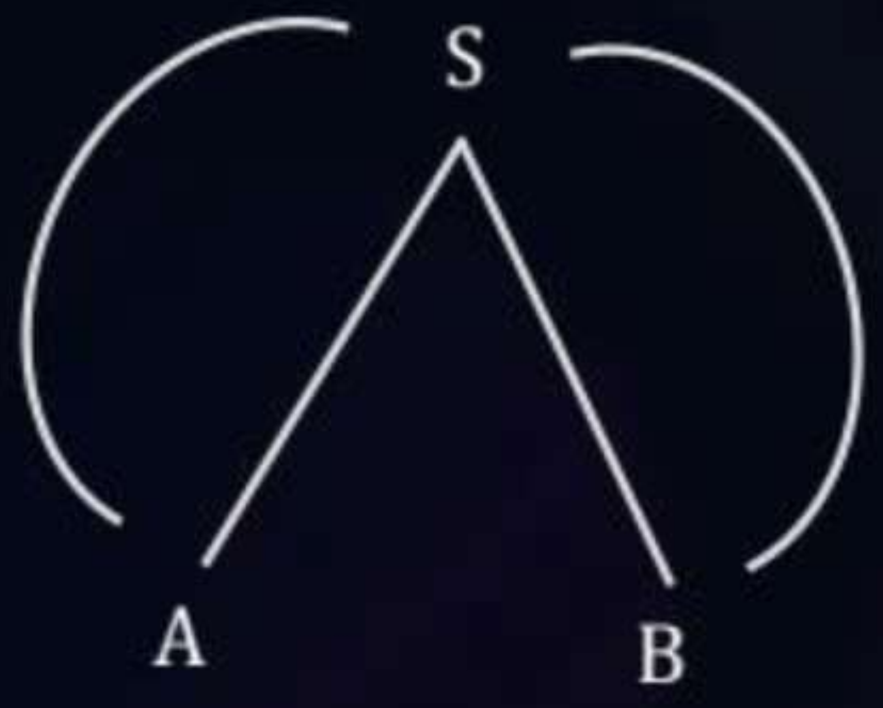


Topic : Syntax Directed Translation

$S \rightarrow AB\ \{A.x = S.x\}$

$A \rightarrow a\ \{A.x = a\}$

$B \rightarrow b\ \{B.x = b\}$



Topic : Syntax Directed Translation

S–attribute Definition:

SDD that use synthesized attribute only

L–attribute Definition:

SDD that use synthesized & Inherited attribute

Topic : Syntax Directed Translation

#Q. $A \to PQ \ \overset{I}{\boxed{\{P.i = A.i\}}} \ \overset{S}{\boxed{\{A.s = Q.i\}}}$

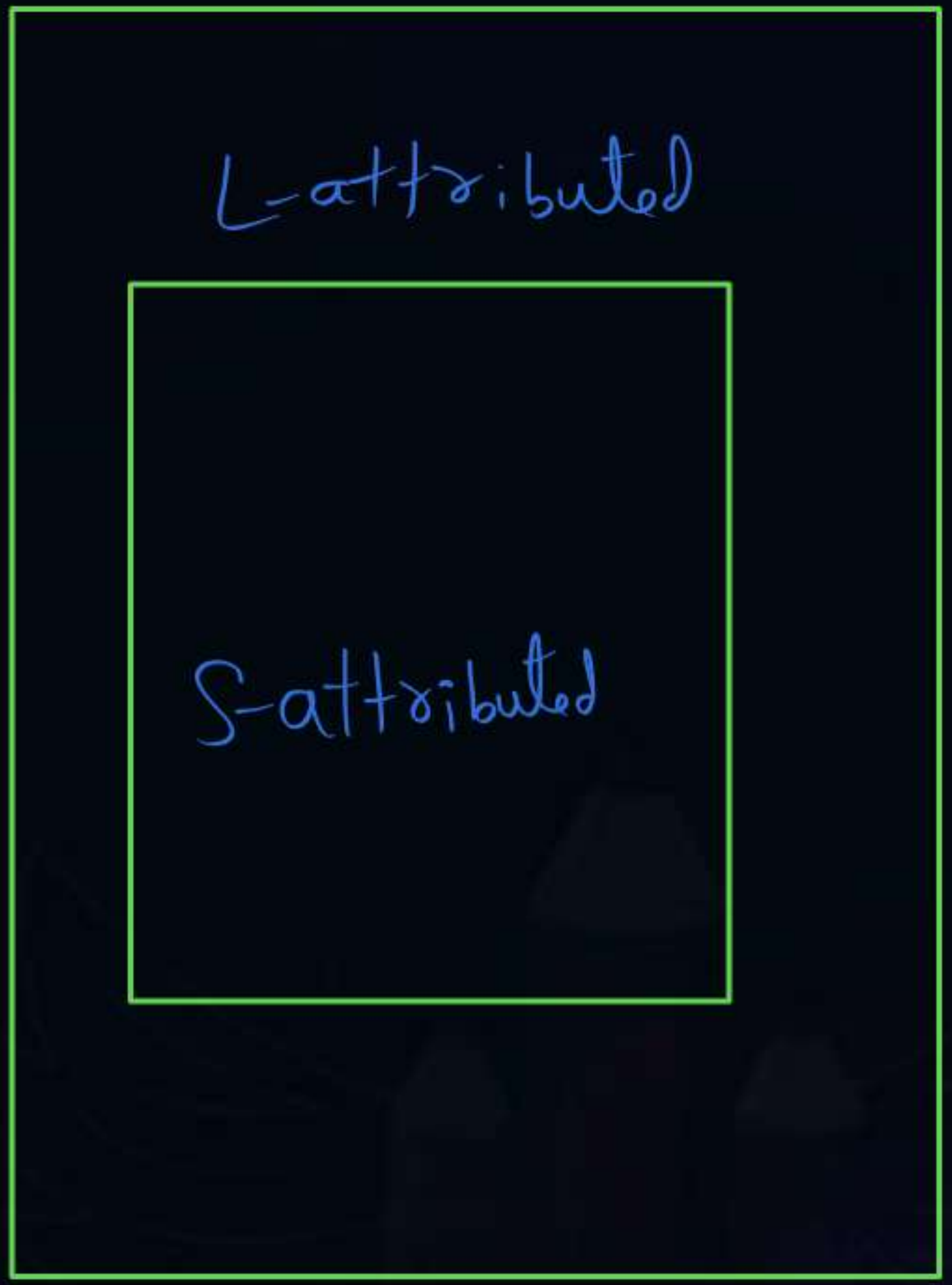
$A \to xy \ \underset{I}{\boxed{\{X.i = A.i\}}} \ \underset{I}{\{Y.i = X.s\}}$

A. S-attributed

B. L-attributed ✓

C. ~~??~~ Both

D. ??



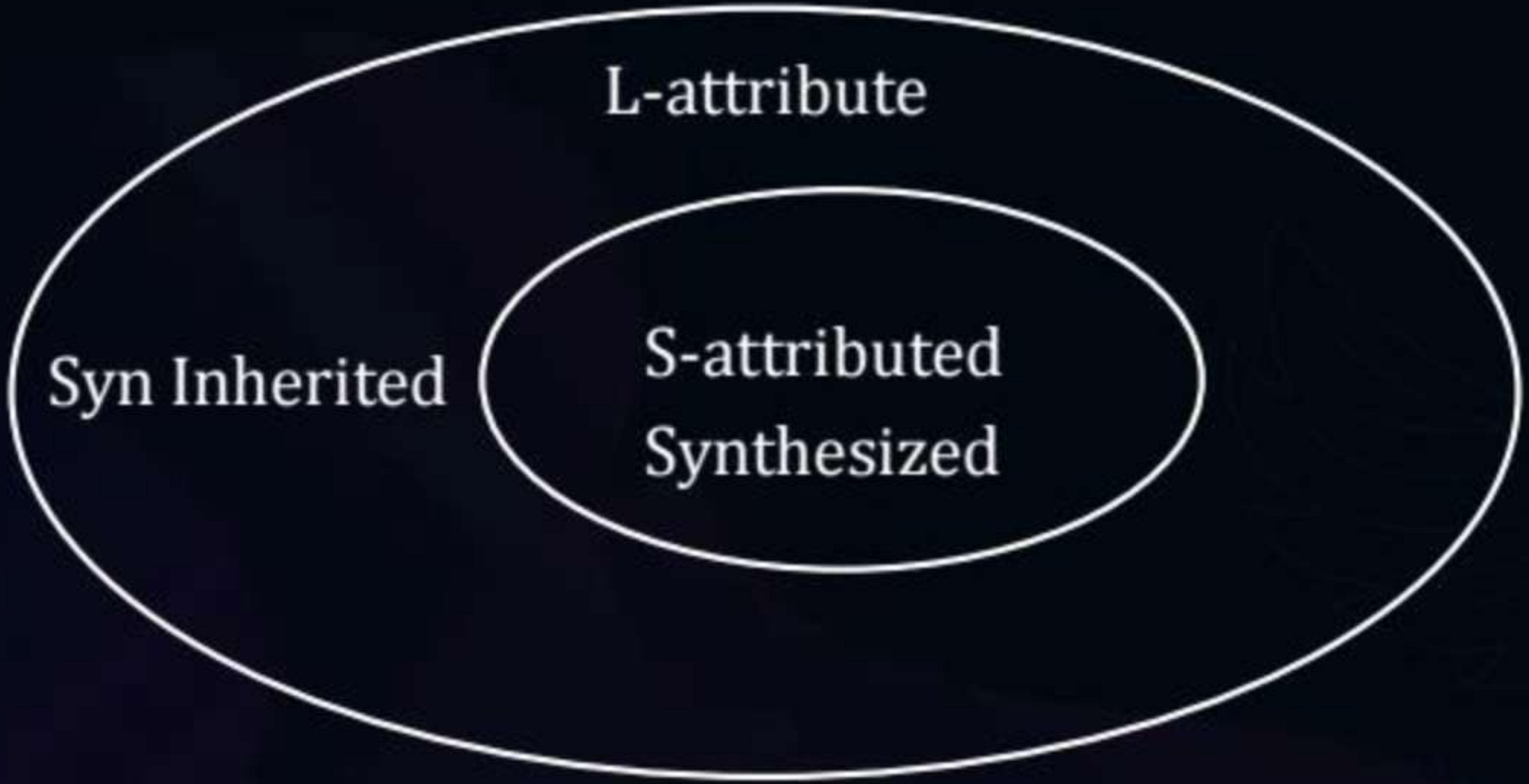
Topic : Syntax Directed Translation

(SDD) S-attribute Definition	L-attribute Definition
(1) Only Synthesized	(1) Both Synthesized & Inherited
(2) Good for Bottom up parsing	(2) Good for top down parsing
(3) Rules evaluation order is Bottom un manner	(3) Rules evaluation order is DFS left to right manner

Topic : Syntax Directed Translation

Every S-attribute is L-attribute

But every L-attributed need not be S-attributed



Topic : Syntax Directed Translation

Translation Scheme

SDT in which semantic rules may present anywhere in R.H.S of grammar production.

Topic : Syntax Directed Translation

{DFS (L-R)}

#Q. Consider the following translation scheme.

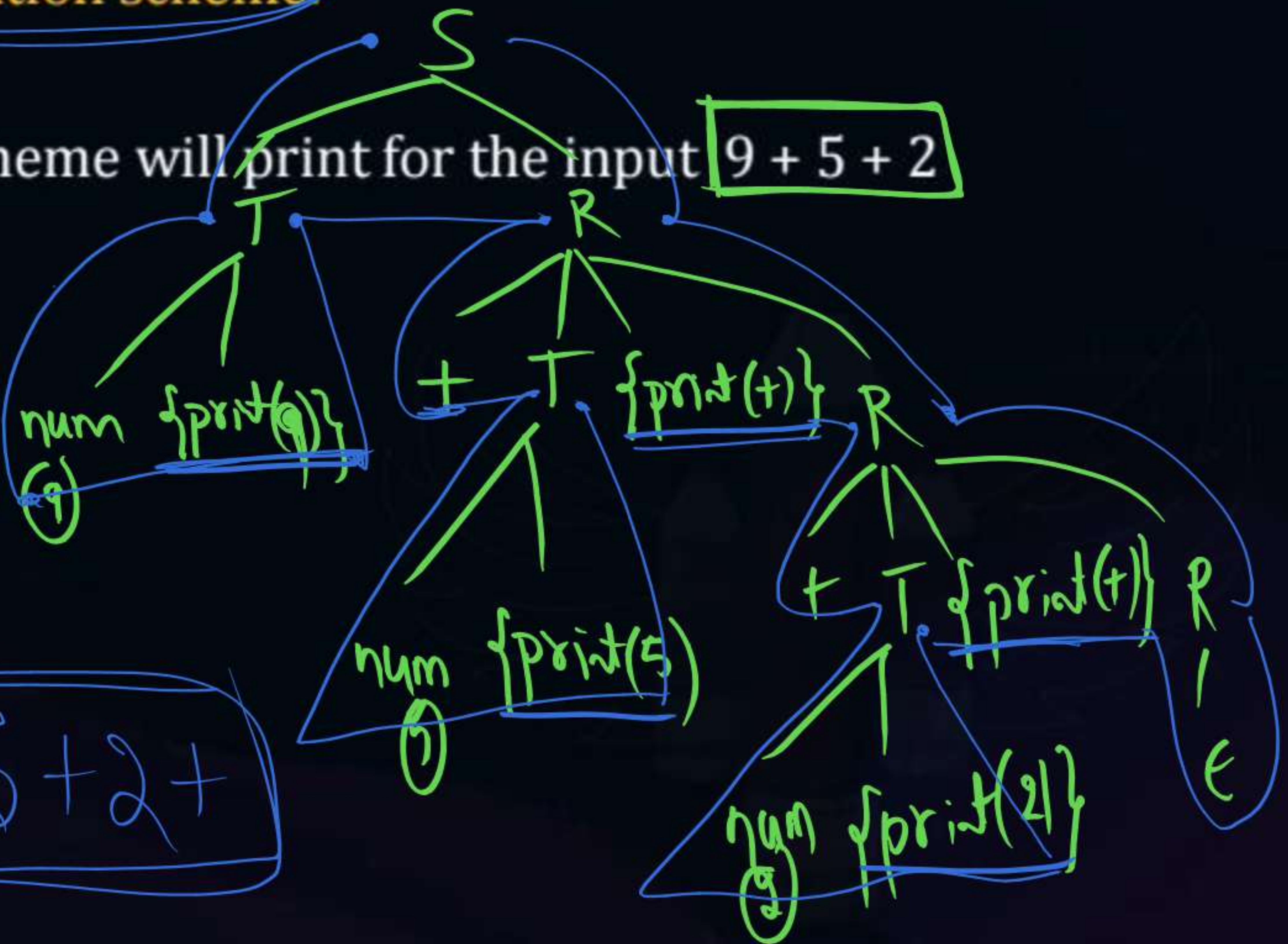
What output this translation scheme will print for the input 9 + 5 + 2

$S \rightarrow TR$

$R \rightarrow +T$ {print (+)} R

$R \rightarrow \epsilon$

$T \rightarrow$ num {print (num. val)}



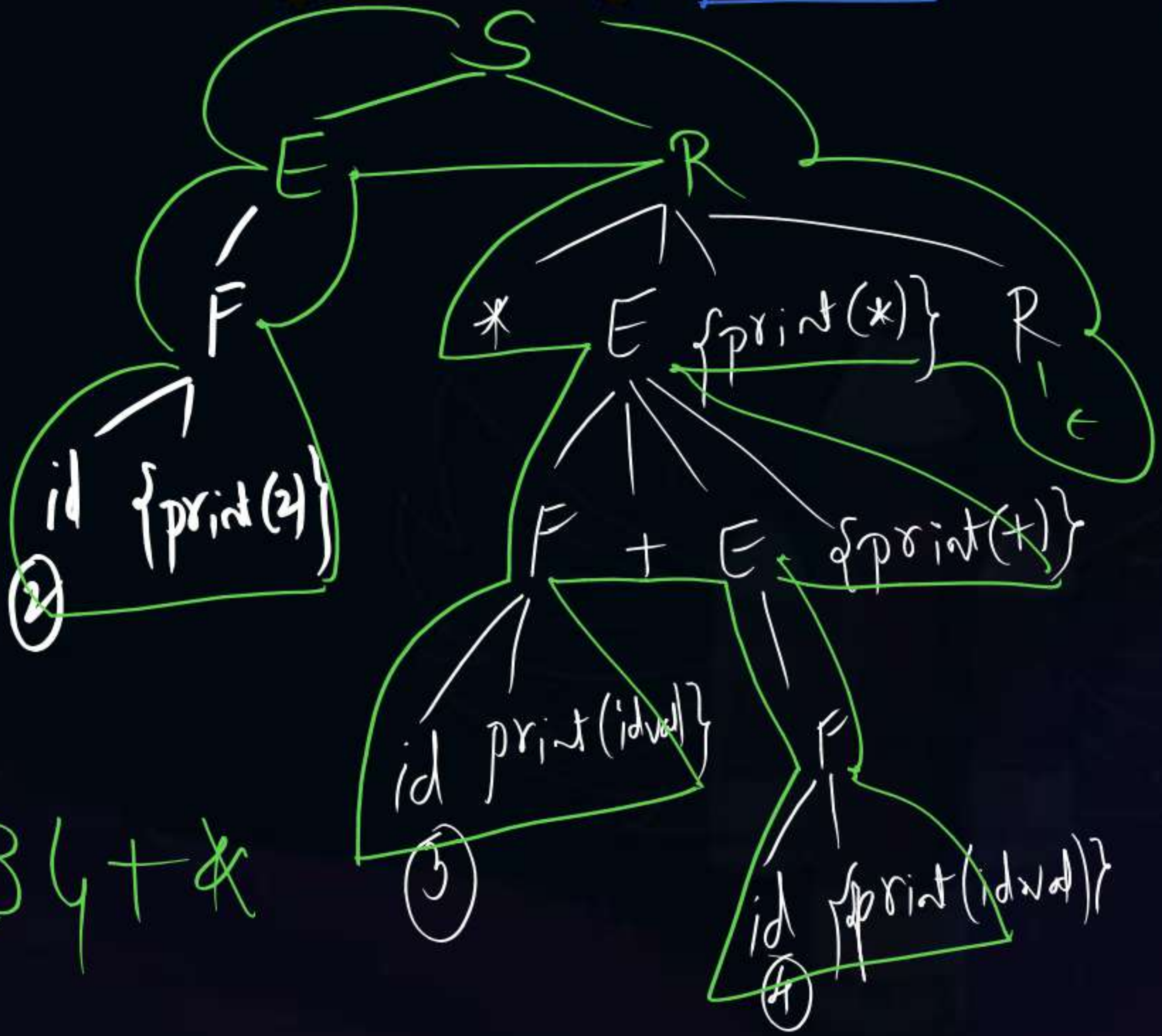
output

95+2+

{DFS (L-R)}

#Q. What output this translation scheme will print for input $\boxed{2*3+4.}$

1) S → ER
2) R → *E {print (*) } R
3) R → ε
4) E → F + E {print (+)}
5) E → F
6) F → S
7) F → id {print (id.val)}



234+*

Topic : Syntax Directed Translation

Home work

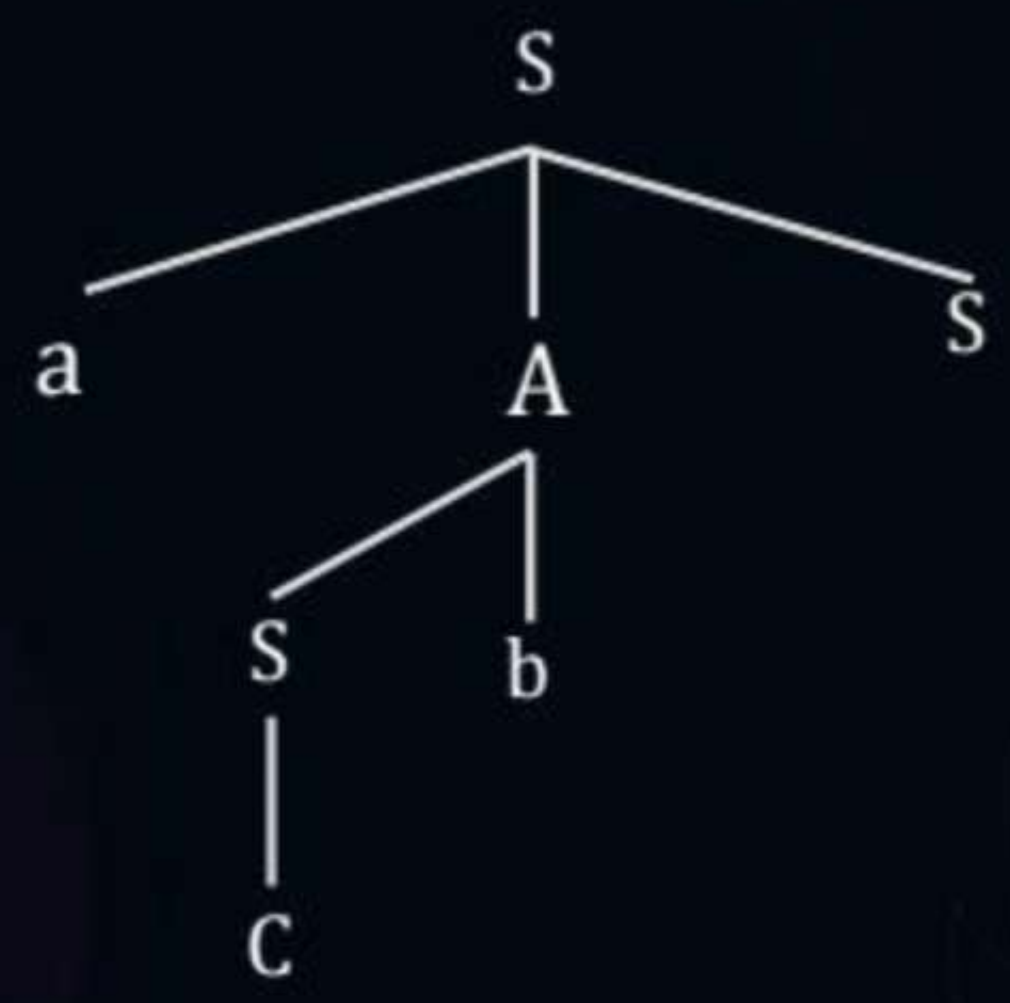
#Q. $S \to aAS$ {print (12)}

$S \to c$ {print (21)}

$A \to Sb$ {print (121)}

What output this translation scheme print for the input $\boxed{\text{acbc.}}$

Topic : Syntax Directed Translation



Topic : Syntax Directed Translation

DFS (L-R)

Home work

#Q. S→ a {print (A)} A

A→ b {print (C)} B

A→ ∈{print (C)}

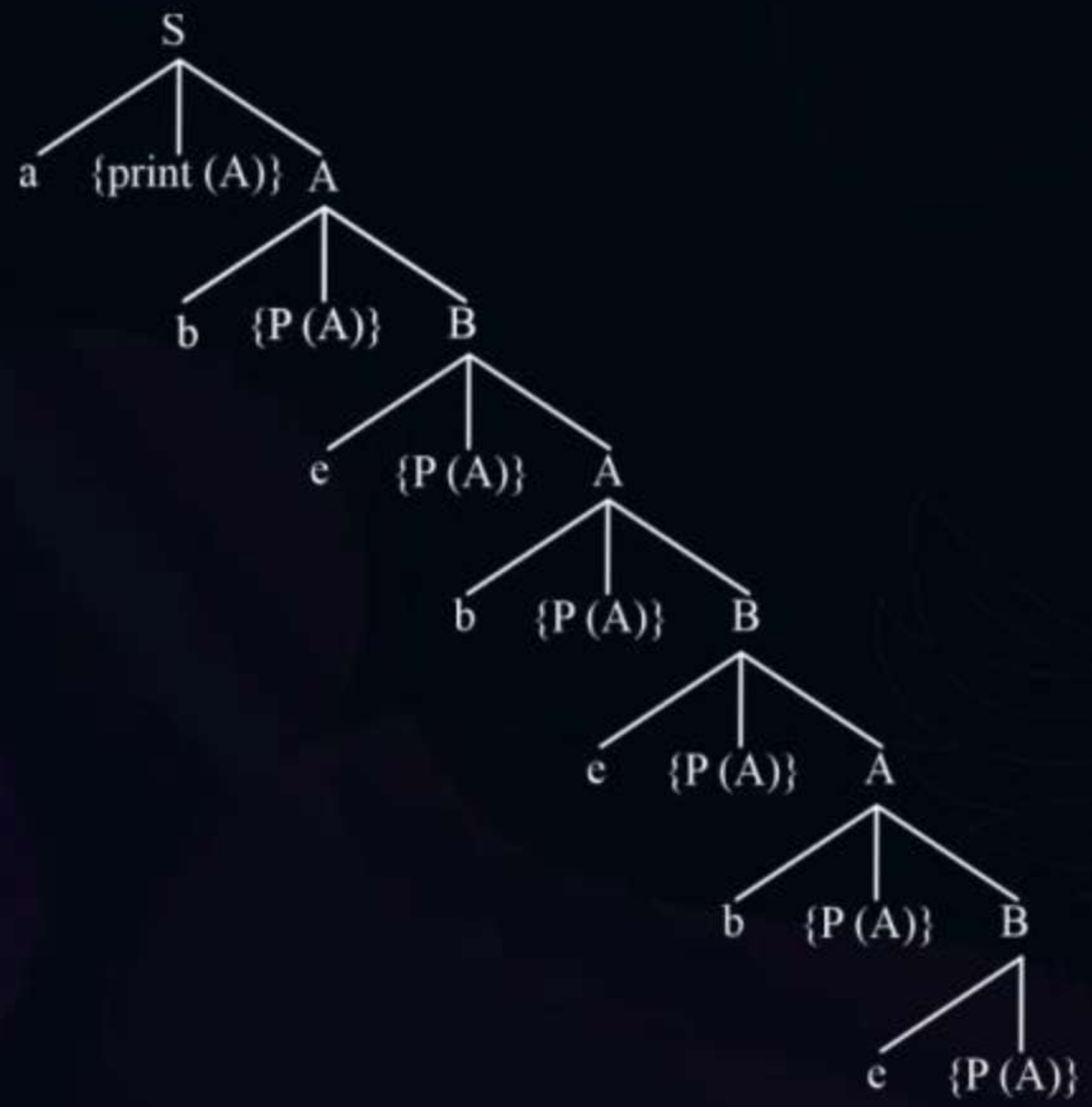
B→ e {print (B)} A

B→ ∈{print (C)}

C→ c {print (A)}

What output above translation scheme for input $\boxed{\text{abebebe}}$

Topic : Syntax Directed Translation



THANK - YOU